Total No. of Questions: 6]

SEAT No. :

P318

[Total No. of Pages : 2

APR-16/BE/Insem. - 22
B.E. (Mechanical)
FINITE ELEMENT ANALYSIS (Elective - IV (b))
(2012 Pattern)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates:

1) ***Draw suitable neat diagrams, wherever necessary.***
2) ***Figures to the right indicate full marks.***
3) ***Use of electronic pocket Calculator is allowed.***
4) ***Assume Suitable data, if required.***

Q1) a) Explain the concept of FEM briefly and outline the procedure. **[6]**

b) Explain the principle of Rayleigh-Ritz method. **[4]**

OR

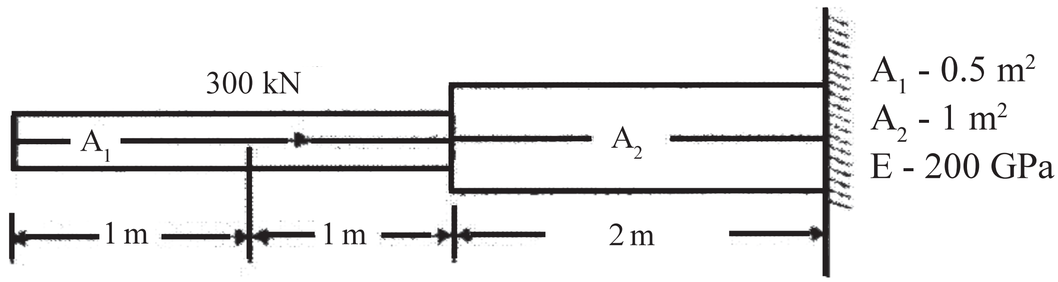
Q2) a) Briefly Outline the concept of P and h formulation and how it is used for improving accuracy in FEA simulations **[6]**

b) Discuss the advantages and disadvantages of FEM over **[4]**

i) Classical method

ii) Finite difference method.

Q3) a) Determine the nodal displacements and Reactions at support by finite element formulation for the following figure. Use P = 300 kN; A, = 0.5m^2; A2 = lm^2,E = 200 Gpa **[6]**

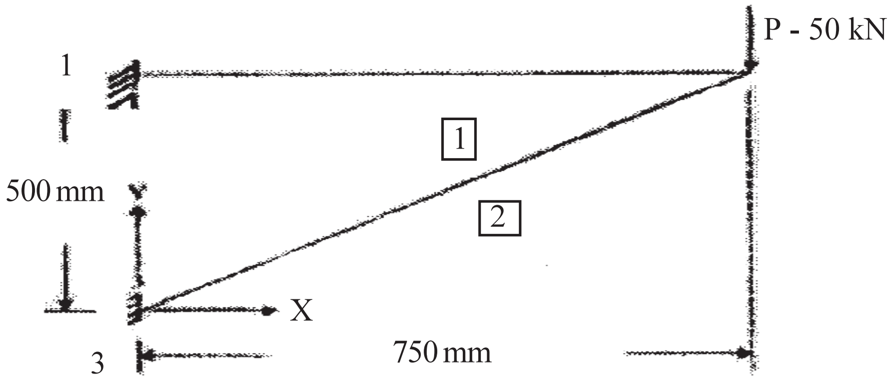


P.T.O.

b) Write a Note on Penalty and Elimination Approach in FEA Solution **[4]**

OR

Q4) Figure below shows a Truss Structure, discretise the model and determine stresses and reactions; assuming points 1 and 3 are fixed Use $E = 2.1 \times 10^5$ N/mm^2 and $A = 1500$ mm^2 **[10]**



Q5) a) Explain what is meant by Plane Stress and how it is used for conversion of 3D problem in 2D problem. Give practical examples **[6]**

b) Write a Note on Linear Strain Triangular Element **[4]**

OR

Q6) a) How Pascal triangle is used to determine a shape function for two-dimensional elements? How rigid body displacement and constant strain rate is taken into account in these polynomial terms? **[6]**

b) What is meant by C0 and C1 type of continuity and how it is taken into account during FEA element formulations? **[4]**